

Manikantan (Mani) Srinivasan

+1 (857)-218-2285

mani.srinivasan2k1@gmail.com

[LinkedIn](#)

[Portfolio](#)

[GitHub](#)

[Medium](#)

EDUCATION

Northeastern University, Boston, Massachusetts

Sep 2023 – Aug 2025
(expected)

Khoury College of Computer Sciences

Master of Science in Artificial Intelligence - **GPA: 3.7/4**

Coursework: NLP, AI in Human Computer Interaction, LLM, Machine Learning, Data Science, Unsupervised ML

Teaching Assistant for CS6220 Data Mining – teaching graduate-level data mining techniques.

UPES, Dehradun, Uttarakhand (India)

Aug 2019 – May 2023

Bachelor of Technology in Computer Science (specialization in AI&ML) – **GPA: 8.3/10**

Coursework: Machine Learning, Algorithms, Deep Learning, Database Management, Software Engineering

SKILLS

Languages:

Python, C++, Java, C, R, HTML, CSS, Bash, MATLAB, Dart, Swift

Software Development Tools

Git, GitLab, Jira, Docker, Kubernetes, Xcode, Tableau, Snowflake

Cloud and Databases

GCP, AWS EC2, SageMaker, S3, ArangoDB, MySQL, MongoDB, Neo4j, Redis, Flutter

Libraries and Frameworks:

PyTorch, HuggingFace, TensorFlow, Django, LangChain, Pandas, Sci-kit Learn, Matplotlib

Domain Skills

Language Modelling, Deep Learning, LLM Fine Tuning, RAG, AI Agents, Data Analytics

PROFESSIONAL EXPERIENCE

SignoFi, Raleigh, USA

Sep 2024 – Jan 2025

AI Research Engineer Intern

- Pioneered an Agentic AI application leveraging Llama 70B and LangChain to retrieve balance sheet data from NoSQL databases, enabling interactive queries and computational analysis for over 50,000 public companies.
- Leveraged RAG and Fine-tuning with adaptive chunking strategies to ground Llama 70B and 90B, boosting the Faithfulness score to **0.84 (84%)** and enhancing contextual precision, recall, and relevancy (**92%**) in factual question answering.
- Architected a custom workflow on Google Cloud Platform using virtual machines, Vertex AI, GCP Buckets, and multi-core GPUs to optimize operations at a portfolio management firm, achieving up to **20%** in cost savings.

KEK, Tsukuba, Ibaraki (Japan)

Oct 2022 – Sep 2023

Grid Computing Developer – R&D

- Collaborated on the Prestigious [BELLE-2](#) particle acceleration experiment as a distributed computing developer. The collaboration involved 5+ world famous laboratories like Brookhaven, CERN, KEK and 127+ institutions like Carnegie Mellon.
- Created innovative software tools that empowered physicists to execute experiments on over 10 PB of data across centers.

Jio, Mumbai, Maharashtra (India)

Jun 2022 – Nov 2022

Software Engineer Intern – (Center of Excellence - AI & ML)

- Engineered the [BrainOS](#) platform integrating 7+ verticals for Reliance Industries, utilizing knowledge graphs and predictive ML across products like JioTV, Hydrocarbons, and JioMart, achieving 86% accuracy and 0.79 precision.
- Developed robust knowledge graphs by preprocessing raw data and implemented 100+ CRUD API functions using gRPC to query NoSQL databases, utilizing debugging tools to analyze and refine 1000+ functions.
- Facilitated the development of a beta version of **collaborative filtering** for product recommendations using **Neo4j**, boosting the product revenue by 10%.

PROJECTS

Machine Unlearning: Erase key concepts from LLM's Knowledge base

- Designed and applied 3 machine unlearning techniques (SAE, LoRA Fine-tuning, Guardrail) on a 7B parameter LLM, ensuring a retained perplexity score under 20 for unaffected knowledge during concept forgetting.
- Performed a comparative analysis with both qualitative and perplexity-based studies. Performed hyper-parameter tuning for PEFT finetune with 1B+ parameters. - <https://github.com/mani2001/Machine-Unlearning>

BlindSight: AI-Powered Voice-Driven File Navigation for the Visually Impaired

- Developed an **AI-powered voice navigation system**, enabling visually impaired users to perform file management and document editing tasks with 95% speech recognition accuracy using Open-AI's C++ implementation of **Whisper**.
- Incorporated **LLaMA-70B** with in-context learning (ICL) to deliver task-specific capabilities, such as document summarization and advanced text editing, achieving 85% accuracy in real-world testing scenarios.
- Designed a multi-threaded architecture that reduced task execution latency by 30%. - <https://github.com/mani2001/BlindSight>

AI Grader assistant: PEFT-Finetuning RoBERTa and BART models for assistive grading

- **Engineered** an automated grading system by fine-tuning RoBERTa (score prediction) and BART (feedback generation) with PEFT LoRA techniques, achieving **0.37 RMSE** and **0.80 Pearson correlation** through optimized training pipelines.
- **Conducted** evaluation of feedback generation using both qualitative and quantitative methods, **tuning hyperparameters** to achieve **0.30 ROUGE-1** and **0.23 ROUGE-L** scores. <https://github.com/mani2001/FineTuningProjectNLP>